B–8–X

Roll No. ....

Total No. of Questions : 22] [Total No. of Printed Pages : 8

12th ARF(SZ)JKUT2024-25
308–X
PHYSICS

Time : 3 Hours] [Maximum Marks : 70

SECTION–A

1 each

1. Choose the correct/most appropriate answer :

   (i) In case of insulators as the temperature decreases, resistivity :

   (A) Increases

   (B) Decreases

   (C) Remains constant

   (D) Becomes zero

   (ii) No force acts on a current carrying conductor in a magnetic field when angle between current and magnetic field is :

   (A) Zero (B) $\frac{\pi}{4}$

   (C) $\frac{\pi}{2}$ (L $\frac{3\pi}{4}$

(iii) Torque acting on a magnet held at angle θ with magnetic field is maximum when θ is equal to :

(A) $0°$

(B) $180°$

(C) $360°$

(D) $90°$

(iv) The SI unit of magnetic flux is :

(A) Weber

(B) Gauss

(C) Orested

(D) Tesla

(v) The dimensions of $\frac{E}{B}$ are same as that of :

(A) Acceleration

(B) Velocity

(C) Charge

(D) Current

(vi) The velocity of light in vacuum is 3×10^8 ms^{-1}. The velocity of light in a medium of refractive index 1.5 is :

(A) 4.5×10^8 ms^{-1}

(B) 3×10^8 ms^{-1}

(C) 2×10^8 ms^{-1}

(D) 1.5×10^8 ms^{-1}

(vii) In a compound microscope, the distance between objective lens and eye lens is :

(A) Fixed

(B) Variable

(C) Infinite

(D) 1 metre

(viii) The series of hydrogen spectrum which lies in visible region is :

(A) Lyman series

(B) Paschen series

(C) Balmer series

(D) None of these

(ix) 'A' stands for atomic mass number and 'Z' for atomic number. The number of electrons in an atom is :

(A) A – Z

(B) A + Z

(C) Z

(D) A

(x) What is the order of the forbidden gap in energy bands of silicon ?

(A) 0.1 eV

(B) 1.1 eV

(C) 2.1 eV

(D) 0.7 eV

SECTION–B

2 each

2. Draw the lines of force due to two equal and similar charges for :

(i) $q_1 = q_2 < 0$ and

(ii) $q_1 = q_2 > 0$

3. An electron is separated from the proton through a distance of 0.53 Å. Calculate the electric field at the location of the electron.

4. State *two* factors on which the sensitivity of a moving coil galvanometer depends.

5. A magnetic flux of 5 μ Wb is linked with a coil when a current of 1 mA flows through it. What is the self-inductance of coil ?

6. State *four* characteristics of e.m. waves.

7. Write *four* applications of optical fibers.

8. What are conditions for sustained interference of light ?

9. Explain the term stopping potential and threshold frequency.

Or

Is photoelectric emission possible at all frequencies ? Give reasons for your answer.

10. What is the de-Broglie wavelength of 0.3 kg object moving with a speed of 6 ms^{-1} ?

SECTION–C 3 each

11. Calculate the number of electrons moving per second through the filament of a lamp of 100 watt operating at 200 volt.

12. Using Kirchhoff's laws derive the condition for balance of a Wheatstone bridge circuit.

Or

Define drift velocity of electrons and establish relation between drift velocity and electric current.

13. A conducting wire of 100 turns is wound over and near the centre of a solenoid of 100 cm length and 2 cm radius having 600 turns. Calculate mutual inductance of two coils.

14. Using phasor treatment derive an expression for impedance and phase angle of LCR circuit.

15. Explain refraction through prism and show $\delta = (i_1 + i_2) - A$.

16. Distinguish between excitation potential and ionization potential.

17. What are nuclear forces ? Give their important characteristics.

18. What is meant by depletion region in a junction diode ? Explain its formation.

19. With the help of a circuit diagram, explain forward characteristics of a *p-n* diode.

SECTION–D

5 each

20. Using Gauss' theorem derive an expression for electric field due to infinitely long straight wire.

Or

What is electric potential ? Derive an expression for electric potential at a distance from a charge +Q.

21. Derive an expression for force acting on a current carrying conductor placed in a uniform magnetic field. Also mention the cases when force is minimum and maximum.

Or

What are dia, para and ferromagnetic substances ? Discuss their important properties.

22. State Huygens' principle and prove the laws of refraction on its basis.

Or

Describe an Astronomical telescope. Drive an expression for its magnifying power when image is formed at infinity.

B–8–X

Roll No. ....

Total No. of Questions : 4] [Total No. of Printed Pages : 7

12th ARM(SZ)JKUT2024
1108–X
PHYSICS

Time : 3 Hours] [Maximum Marks : 70

SECTION–A

1 each

1. (i) A body can be negatively charged by :

(A) Giving excess of electrons to it

(B) Removing some electrons from it

(C) Giving some protons to it

(D) Removing some neutrons from it

(ii) It is possible to have a positively charged body at :

(A) Zero potential

(B) Negative potential

(C) Positive potential

(D) All of these

(iii) Which of the following relation is called as current density ?

(A) $\frac{I}{A}$

(B) $\frac{A}{I}$

(C) $\frac{I^2}{A}$

(D) $\frac{I^3}{A^2}$

(iv) Which of the following is ferromagnetic ?

(A) Aluminium

(B) Quartz

(C) Nickel

(D) Bismuth

(v) A transformer is used to change

(A) High voltage d.c. into low voltage d.c.

(B) High voltage a.c. into low voltage a.c.

(C) Electrical energy into mechanical energy

(D) Mechanical energy into electrical energy

(vi) Which of the following are not electromagnetic wave

(A) Cosmic rays

(B) γ-rays

(C) β-rays

(D) X-rays

(vii) Which of the following produces a plane wave front ?

(A) Point source

(B) Line source

(C) Extended source

(D) None of these

(viii) Electron volt is the unit of :

(A) Charge

(B) Momentum

(C) Potential difference

(D) Energy

(ix) In case of artificial ıadioactive transformation as given by :

$$_{15}P^{30} \rightarrow {}_{14}Si^{30} + X,$$

particle X is :

(A) Neutron

(B) Proton

(C) Electron

(D) Positron

(x) The type of bond in a silicon crystal is :

(A) Ionic

(B) Metallic

(C) Covalent

(D) Van der Waals

SECTION–B

2 each

2. (a) What is an equipotential surface ? Give the direction of electric field with respect to an equipotential surface.

(b) Self induction is called the inertia of electricity. Explain. why.

(c) What is Maxwell's displacement current ? Is displacement current, a source of magnetic field ?

(d) When does Snell's law of refraction fail ?

(e) For glass-air interface, the critical angle is C. Will the critical angle for glass-water interface be greater or less than C ? Explain.

(f) A ray of light incident on an equilateral glass prism ($\mu_g = \sqrt{3}$) moves parallel to the base line of the prism inside it. Find the angle of incidence for this ray.

(g) Why do we fail to observe the diffraction from a wide slit illuminated by monochromatic light ?

(h) What are nuclear forces ? State their three properties.

(i) Write two important significances of binding energy per nucleon curve.

SECTION–C

3 each

3. (a) Derive a relation between e.m.f., potential difference and internal resistance of a cell.

(b) A 10 C charge flows through a wire in 5 minutes. The radius of the wire is 1 mm. It contains 5×10^{22} electrons per centimeter^3. Calculate current and drift velocity.

(c) What is an ammeter ? How can a galvanometer be converted into an ammeter ?

(d) State Lenz's law and show that it does not violate the law of conservation of energy.

(e) A capacitor of 1 μF is connected to source of a.c. having e.m.f. given by equation $E = 200 \cos 120\, \pi t$. Find r.m.s. value of current through the capacitor.

(f) Calculate de-Broglie equation for a material particle.

(g) On the basis of Bohr's atomic model, find an expression for radius of *n*th orbit of a hydrogen atom.

(h) Explain the forward and reverse bias characteristic curve of a *p-n* junction.

(i) Distinguish between intrinsic and extrinsic semiconductors.

SECTION–D

5 each

4. (a) State Gauss's law in electrostatics. Derive an expression for the electric field due to an infinitely long straight charged wire at a point distant r from it. Plot a graph showing the variation of electric field with r.

Or

What is a capacitor ? Derive an expression for total capacitance when three capacitors of capacitances C_1, C_2 and C_3 are connected in (i) series (ii) parallel.

(b) What is magnetic dipole and magnetic field intensity ? Derive an expression for the torque acting on a bar magnet placed in a uniform magnetic field.

Or

Derive an expression for the force per unit length experienced by each of the two long current carrying conductors placed parallel to each other in air. Hence, define one ampere of current.

(c) What is interference of light ? Deduce the conditions for constructive and destructive interference in Young's double slit experiment.

Or

What is compound microscope ? With the help of ray diagram, explain the working of compound microscope. Find an expression for its magnifying power.

D–14–A

Roll No. ....

Total No. of Questions : 26]

[Total No. of Printed Pages : 7

12^{th}ARJKLK23
9614–A
PHYSICS

Time : 3 Hours]

[Maximum Marks : 70

SECTION–A

(VERY-VERY SHORT ANSWER TYPE QUESTIONS) 1 each

1. Of metals and alloys which have greater temperature coefficient of Resistance ?

2. Which has smaller wavelength microwaves or infra-red waves ?

3. Name the spectral series of hydrogen atom having least wavelength.

4. How will photocurrent change on decreasing wavelength of incident light ?

5. Define Doping of a Semiconductor.

SECTION–B

(VERY SHORT ANSWER TYPE QUESTIONS) 2 eacl

6. Define mutual induction and coefficient of mutual inductance.

Or

A 5 μF capacitor is connected to a 50 Hz a.c. supply. Calculate its capacitive reactance.

7. Give some important uses of U-V rays.

8. Define dispersive power of prism. Write is formula.

9. What are Isotopes ? Give some examples.

10. Define Frequency Modulation.

SECTION–C

(SHORT ANSWER TYPE QUESTIONS) 3 each

11. Find an expression for the energy stored in a capacitor.

Or

Explain superposition principle of forces.

12. Discuss variation of resistance with temperature and hence define temperature coefficient of resistance.

13. A battery of emf 10 V and internal resistance 3 Ω is connected to a resistor. If the current in circuit is 0.5 A, what is resistance of resistor ? What is the terminal voltage of battery when circuit is closed ?

14. Two moving coil galvanometers M_1 and M_2 having the following particulars :

$R_1 = 10\,\Omega$, $N_1 = 30$, $A_1 = 3.6 \times 10^{-3}\,m^2$, $B_1 = 0.25\,T$

$R_2 = 14\,\Omega$, $N_2 = 42$, $A_2 = 1.8 \times 10^{-3}\,m^2$, $B_2 = 0.50\,T$

(The spring constants are same for two meters).

Determine ratio of :

(i) Current Sensitivity

(ii) Voltage Sensitivity of M_2 and M_1

15. What are Eddy Currents ? Give some important applications of eddy currents.

16. What is meant by resonance in A.C. circuits ? Derive expression for resonant frequency in series L.C.R. circuit.

17. A ray of light suffers minimum deviation, while passing through a prism of refractive index 1.5 and refracting angle of 60°. Calculate angle of minimum deviation and angle of incidence.

18. Using Einstein's Photoelectric Equation explain laws of photoelectric effect.

19. State radioactive decay law and hence obtain equation $N = N_0 e^{-\lambda t}$.

20. Draw circuit diagram of full wave rectifier. Explain its working and draw its input and output waveforms.

21. What is AND gate ? Write its symbol Boolean expression and truth-table.

22. Explain sky wave propagation and space wave propagation.

SECTION–D

(VALUE BASED QUESTION) 4

23. Sushma decided to give a camera as gift to her younger brother on his birthday. She visited a shop for purchasing camera. The shopkeeper showed her some cameras and advocated a beautiful camera having power of lens equal to that of another high priced camera. She asked shopkeeper as to why this beautiful camera was low priced. The shopkeeper told her an equivalent lens has been used in camera. He said that he judged her to be a student and thought to suggest low budget camera to be good option. Sushma being science student happily purchased the camera. Now answer the following questions :

(a) What is an equivalent lens ?

(b) What is the relation of power of equivalent lens ?

(c) What do you think about the moral values of shopkeeper ?

SECTION–E

(LONG ANSWER TYPE QUESTIONS) 5 each

24. Define electric potential energy. Derive an expression for electric potential energy of a system of two point charges.

Or

Explain the principle on which Vandegraff generator works ? Draw labelled diagram and describe its construction and working.

25. What is Cyclotron ? Discuss its construction, working and theory. What are its limitations ?

Or

State Ampere Circuital Law. Derive an expression for the magnetic field due to a straight current carrying solenoid.

26. What is an Astronomical Telescope ? Draw course of rays through it and hence derive expression for magnifying power in it when final image is formed at least distance of distinct vision.

Or

Derive an expression for fringe-width in Young's double slit experiment.

A–1–A

Roll No.................................

Total No. of Questions : **26**] [Total No. of Printed Pages : **4**

12^{th} SZARJD22
6001–A
PHYSICS

Time : 2.30 Hours] **[Maximum Marks : 70**

Note :– Attempt all questions.

Section–A

1 each

1. Charge resides only on the outer surface of a charged conductor. (True/False)

2. What is the condition of balanced position of Wheatstone bridge ?

3. Ozone layer is present in the stratosphere region of atmosphere. (True/False)

4. Define threshold frequency.

5. P-type semiconductor is electrically neutral. (True/False)

Section–B

2 each

6. Find the electrostatic potential energy of an electric dipole having magnitude of each charge as 3×10^{-6} coulomb, separated by a distance of 2×10^{-7} metre.

Or

Three capacitors of capacitance 1μF, 2μF and 3μF are connected in series. Find the total capacitance of combination.

7. Mention two points of difference between step up and step down transformer.

8. Give *four* properties of electromagnetic waves.

9. Find the magnifying power and length of astronomical telescope of objective and eye piece of focal length 144 cm and 6 cm respectively.

10. Why ground waves are not suitable for high frequency ?

Section–C 3 each

11. Derive relation between Current and Drift velocity.

Or

Find the resistivity of wire of length 2 m, diameter 0.01 m and resistance 50 mΩ.

12. A battery of e.m.f. 10 V and internal resistance 3 Ω is connected to a resistor so that a current of 0.5 A flows in the circuit. Find the resistance of resistor and terminal voltage of battery, in a closed circuit.

13. A solenoid is 2 m long and 3 cm in diameter, has 5 layers of winding of 1000 turns, each carries a current of 5 A. Find magnetic induction at its centre along its axis and also at the ends.

14. State and explain Faraday's laws of electromagnetic induction.

15. Magnetic field through a coil of 200 turns and area of cross-section *0.04* m^2 *changes from* 0.1 wb/m^2 to 0.04 wb/m^2 in 0.02 second. Find the induced e.m.f. and induced current, if resistance of coil is 10 ohm.

16. Define Scattering of light. Why sky appears blue ?

17. Derive Einstein's photoelectric equations.

18. Write down the postulates of Bohr's model of hydrogen atom.

19. Define nuclear fission. Explain with the help of an example.

20. Distinguish between *p*- and *n*-type extrinsic semiconductor.

21. Give the truth table, logic symbol and Boolean expression of OR gate.

22. Draw the block diagram of data transmission and data reception.

Section–D 4

23. Owais was very happy to receive a car on his birthday from his father. Then Owais and his friend Basit went for a long drive at night. Basit suggested Owais to get the tubes of his headlight replaced with high power tubes, as the headlight according to him were weak. Owais did so and again went for long drive with his friend to check the new headlights. As they were enjoying the drive, suddenly a scooter coming from opposite side struck their car. Luckily nobody was hurt but the vehicles got badly damaged. Basit slapped the scooterist but the scooterist said that the headlights of the car made him blind and he could not see anything. Owais's father became very angry with Owais and told him that he should have installed special cover on the headlights to reduce the glare. Now, answer the following :

   (a) What is the special cover reducing the glare called ? Mention its two other applications.

(b) What is your opinion about Owais and Basit's behaviour ?

(c) What do you think about the reaction of Owais's father ?

(d) Name and state the law that relates refractive index of the material cover with the angle at which reflected light and refracted light are perpendicular to each other.

Section–E

5 each

24. State Gauss' law. Derive an expression for the electric field due to an infinite line of charge.

Or

Give the principle, construction and working of Van de Graff's generator.

25. State Biot Savart's law. Derive an expression for the magnetic field at the centre of circular coil carrying current.

Or

What are diamagnetic, paramagnetic and ferromagnetic substances ? Give their properties.

26. State Huygens' principle. Derive laws of reflection or refraction from it.

Or

What is Lens Maker's formula ? Derive an expression for Lens Maker's formula for a convex lens.

1901-Z

PHYSICS

Maximum Marks—70

Time Allowed—3 Hours
(Fifteen Minutes Extra to read the Question Paper)

(Long Answer Type Questions)

1. State and explain Coulomb's law in vector form. Hence defineunit charge.

Or

Show that electric potential can be represented as line integral of electric field. 5

2. Find the force between two Parallel conductors carrying current in Same and Opposite direction and hence define one ampere.

Or

What is Ampere circuit law? Derive an expression for magnetic field due to a current in a toroid. 5

3. State and explain Faraday's laws of Electromagnetic induction.

Or

What is a Transformer? Explain its theory and discuss its main uses. 5

4. What do you mean by total internal reflection and critical angle? Derive a relation between the refractive index and critical angle. What are the conditions for producing total internal reflection?

Or

With the help of ray diagram, explain the formation of image in compound microscope. Derive an expression for its magnifying power. 5

(Short Answer Type Questions)

5. How is a Potentiometer used to compare e.m.f. of two cells? 3

6. State Ohm's law and deduce it from the knowledge of drift velocity. 3

7. Three capacitors of capacitance 2PF, 3PF and 4PF are connected in parallel. What is the charge on each capacitor of the combination is connected to 100 volt supply? 3

8. What is meant by Electric resonance? Where is it used?

9. What does an Electromagnetic wave convert of ? On what factors does its velocity in vacuum depend? 3

10. Distinguish between Interference and Diffraction. 3

11. Show that in case of a prism $\hat{A} + \hat{D} = \hat{i} + \hat{e}$ where symbols have their usual meanings. 3

12. State difference between Nuclear fusion and Fission. Also give an example of each.

(Very Short Answer Type Questions)

13. The following very short answer type questions of two marks, each may be answered in a few words or few sentences or as may be required.

(a) State two properties of Paramagnetic substance. 2

(b) Distinguish between Constructive and Destructive interference. 2

(c) What is Threshold frequency and Stopping potential?

(d) What is the Ionization energy of H-atom? 2

(e) Draw the circuit diagram of a pnp transistor as an amplifier.

(f) How can NOR gate be made? Write its Boolean expression. 2

(g) What is Amplitude modulation?

(h) What is Sky-wave? Where it is used?

(Objective Type Questions)

14. (i) Which alloys are used for making standard resistance coils? 1

(ii) What is the magnitude of Force acting on a stationary charge? 1

(iii) The speed of light in a medium is independent of the nature of source, why? 1

(iv) What is the velocity of a Photon? 1

(v) How many electron volt make one joule? 1

Choose the correct answer :

(vi) The density of nucleus is of the order of

A. $10^{3} kg/m^{3}$

B. $10^{10} kg/m^{3}$

C. $10^{15} kg/m^{3}$

D. $10^{17} kg/m^{3}$. 1

(vii) The impurity atoms with which pure silicon should be dropped to make P-type semiconductor

A. Phosphorus

B. Antimony

C. Boron

D. Arsenic.